Clase 5: Regresión logística Cont.

18/Marzo.

$$\ell(y,p) = -[y \log p + (1-y)\log(1-p)]$$

$$\boxed{\nabla_\beta \ell(y, P_\beta(x)) = ?}$$

i) $\frac{\partial}{\partial z}\sigma(\overset{\in\mathbb{R}}{z}) = \sigma(z)(1-\sigma(z))$

ii) $\frac{\partial}{\beta_j}\sigma(\underset{\mathbb{R}^{d+1}}{\beta^T x}) = \underbrace{x_j}_{\in\mathbb{R}}\,\sigma(\beta^T x)(1-\sigma(\beta^T x))$

iii) $\nabla_\beta \sigma(\underset{\mathbb{R}^{d+1}}{\beta^T x}) = \underset{\in\mathbb{R}^{d+1}}{x} \cdot \underbrace{\sigma(\beta^T x)(1-\sigma(\beta^T x))}_{\in\mathbb{R}}$

iv) $\frac{\partial}{\partial\beta_j}\ell(y, P_\beta(x)) = \frac{\partial}{\partial\beta_j}\left\{-[y\log P_\beta(x) + \underbrace{(1-y)\log(1-P_\beta(x))}]\right\}$; $P_\beta(x) = \sigma(\beta^T x)$

$$= -y\,x_j\,\frac{\cancel{\sigma(\beta^T x)}(1-\sigma(\beta^T x))}{\cancel{\sigma(\beta^T x)}} + (1-y)\frac{x_j\,\sigma(\beta^T x)\cancel{(1-\sigma(\beta^T x))}}{\cancel{1-\sigma(\beta^T x)}}$$

$$= -y x_j(1-\sigma(\beta^T x)) + x_j\,\sigma(\beta^T x) - y\cdot x_j\,\sigma(\beta^T x)$$

$$= \underline{-y x_j} + y x_j\,\cancel{\sigma(\beta^T x)} + \underline{x_j\,\sigma(\beta^T x)} - y\,x_j\,\cancel{\sigma(\beta^T x)}$$

$$= x_j(\sigma(\beta^T x) - y) \in \mathbb{R}$$

v) $\nabla_\beta \ell(y, P_\beta(x)) = \underset{\mathbb{R}^{d+1}}{x}\underbrace{(\sigma(\beta^T x) - y)}_{\mathbb{R}}$

Ahora con datos.

$$\hat{R}(\beta) = \frac{1}{n}\sum_{i=1}^{n}\underbrace{\ell(y_i, X_i)}$$

Recordatorio!

$$X = \begin{pmatrix} - & X_1 & - \\ - & X_2 & - \\ & \vdots & \\ - & X_n & - \end{pmatrix} \qquad y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$\nabla_\beta \hat{R}(\beta) = \frac{1}{n}\sum_{i=1}^{n} X_i^T(\underbrace{\sigma(X_i\beta)}_{p_i} - y_i)$$

$$= \frac{1}{n} \sum_{i=1}^{n} \underbrace{\underset{\mathbb{R}^{d+1}}{x_i^T} \underbrace{(p_i - y_i)}_{\mathbb{R}}}$$

$$= \frac{1}{n} \begin{pmatrix} | & & | & & | \\ (p_1 - y_1)\, x_1 & + & (p_2 - y_2)\, x_2 & + \cdots + & (p_n - y_n)\, x_n \\ | & & | & & | \end{pmatrix}$$

Acá está β.

$$= \frac{1}{n} \underset{(d+1)\times n}{X^T} (p - y)_n$$

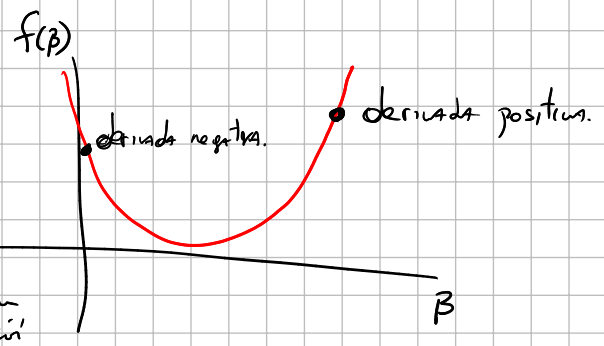
Ahora nos gustaría igualar a 0 y resolver β.

$$X^T p = X^T y$$

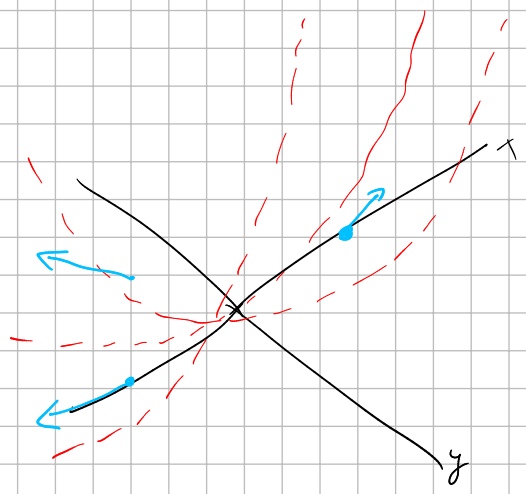
$$\Rightarrow X^T \begin{pmatrix} \sigma(x_1 \beta) \\ \sigma(x_2 \beta) \\ \vdots \\ \sigma(x_n \beta) \end{pmatrix} = X^T \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

II. Gradient Descent

Idea en $\mathbb{R}$.



Ir contra la derivada = achicar la función



Ir contra el gradiente achica el valor de la función.

Matemáticamente, cómo vamos contra el gradiente?

tenemos la función $\hat{R}(\beta)$

$$\nabla_\beta \hat{R}(\beta) = \frac{1}{n} X^T (p - y)$$

$$p \to \begin{pmatrix} \sigma(x_1 \beta) \\ \sigma(x_2 \beta) \\ \vdots \\ \sigma(x_n \beta) \end{pmatrix}$$

La idea de GD es empezar en un valor de $\beta \in \mathbb{R}^{d+1}$; $\beta^{(0)}$

Así vamos contra el gradiente:

$$\beta^{(1)} = \beta^{(0)} - \eta \left(\nabla_\beta \hat{R}\right)(\beta^{(0)})$$

$\eta \to$ tamaño de salto.

iteramos

$$\beta^{(2)} = \beta^{(1)} - \eta \left(\nabla_\beta \hat{R}\right)(\beta^{(1)})$$

$$\beta^{(3)} = \beta^{(2)} - \eta \left(\nabla_\beta \hat{R}\right)(\beta^{(2)})$$

$$\beta^{(i+1)} = \beta^{(i)} - \eta \left(\nabla_\beta \hat{R}\right)(\beta^{(i)})$$

si pudiesemos tomar límite $\lim_{i \to \infty}$

$$\cancel{\beta_L} = \cancel{\beta_L} - \eta \left(\nabla_\beta \hat{R}\right)(\beta_L)$$

$$\Rightarrow 0 = \nabla_\beta (\hat{R})(\beta_L) \leftarrow \text{punto crítico}$$